📘 **AI/ML Internship – Day 41 Notes & Tasks**

🟢 **Module 7: Random Forest Classification Algorithm**

📌 **Part 1: Introduction to Random Forest**

🔹 **What is Random Forest?**

👉 Random Forest is a Supervised Machine Learning algorithm used for:

- Classification
- Regression

👉 It is made up of multiple Decision Trees working together.

🔹 **Why is it called Random Forest?**

👉 A forest contains many trees.

Similarly,

👉 Random Forest contains many Decision Trees.

Tree 1
Tree 2
Tree 3
Tree 4
Tree 5
 ↓
Final Decision

📌 **Part 2: Real-Life Example**

Imagine you want to buy a mobile phone.

Instead of asking one person:

❌ Ask only one friend

You ask:

✅ 10 friends

Then choose the answer that most people suggest.

Random Forest works the same way.

Multiple trees vote and the majority answer becomes the final prediction.

📌 Part 3: Theory Answers (IMPORTANT)

🔹 1. What is Random Forest?

👉 Random Forest is an ensemble machine learning algorithm that combines multiple Decision Trees to improve prediction accuracy.

🔹 2. Is Random Forest Supervised or Unsupervised?

👉 Random Forest is a Supervised Learning algorithm.

🔹 3. What is an Ensemble Method?

👉 Ensemble means combining multiple models to make better predictions.

🔹 4. Why is Random Forest Better than a Single Decision Tree?

👉 Because:

- Higher accuracy
- Less overfitting
- More reliable predictions

🔹 5. What is Voting in Random Forest?

👉 Each tree gives a prediction.

The majority prediction becomes the final answer.

📌 Part 4: How Random Forest Works

Step 1

Create multiple Decision Trees.

Step 2

Train each tree using different data samples.

Step 3

Each tree makes a prediction.

Step 4

Majority voting determines final prediction.

📌 Part 5: Visualization

```
          Random Forest

     Tree1 → Pass
     Tree2 → Pass
     Tree3 → Fail
     Tree4 → Pass
     Tree5 → Pass

        Majority Vote

          PASS
```

📌 Part 6: Example Dataset

Study Hours	Result
1	Fail
2	Fail
3	Fail
5	Pass

Study Hours	Result
6	Pass
7	Pass

📌 Part 7: Import Libraries

```
import pandas as pd

from sklearn.ensemble import RandomForestClassifier
```

📌 Part 8: Create Dataset

```
data = {
   "Hours":[1,2,3,5,6,7],
   "Result":[0,0,0,1,1,1]
}

df = pd.DataFrame(data)

print(df)
```

Meaning

Value	Meaning
0	Fail
1	Pass

📌 Part 9: Features and Labels

```
X = df[["Hours"]]

y = df["Result"]
```

📌 Part 10: Create Random Forest Model

model = RandomForestClassifier()

📌 Part 11: Train the Model

model.fit(X, y)

📌 Part 12: Make Prediction

prediction = model.predict([[4]])

print(prediction)

Possible Output:

[1]

Meaning:

👉 PASS

📌 Part 13: Predict Multiple Values

prediction = model.predict([[2],[5],[8]])

print(prediction)

Possible Output:

[0 1 1]

📌 Part 14: Complete Program

```
import pandas as pd
from sklearn.ensemble import RandomForestClassifier

data = {
  "Hours":[1,2,3,5,6,7],
  "Result":[0,0,0,1,1,1]
}
```

```
df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Result"]

model = RandomForestClassifier()

model.fit(X, y)

prediction = model.predict([[5]])

print("Prediction:", prediction)
```

📌 Part 15: Important Parameters

Number of Trees

```
model = RandomForestClassifier(
    n_estimators=100
)
```

Meaning

n_estimators = Number of Trees

Example:

- 10 Trees
- 50 Trees
- 100 Trees

More trees generally improve stability.

📌 Part 16: Advantages of Random Forest

✅ High Accuracy

✅ Reduces Overfitting

✅ Handles Large Datasets

✅ Reliable Predictions

✅ Works with Classification & Regression

📌 Part 17: Disadvantages of Random Forest

❌ Slower than a single Decision Tree

❌ Uses more memory

❌ Harder to visualize

📌 Part 18: Real-World Applications

Banking

👉 Loan Approval

Healthcare

👉 Disease Prediction

E-Commerce

👉 Customer Classification

Security

👉 Fraud Detection

Education

👉 Student Performance Prediction

📌 Part 19: Decision Tree vs Random Forest

Decision Tree	Random Forest
Single Tree	Multiple Trees
Can overfit	Less overfitting
Faster	Slightly slower
Less accurate	More accurate

📌 Part 20: Random Forest Workflow

Dataset
↓
Multiple Trees
↓
Individual Predictions
↓
Majority Voting
↓
Final Prediction

📌 Part 21: Why Random Forest is Popular?

👉 Many real-world AI systems use Random Forest because:

- Accurate
- Stable
- Easy to use
- Handles large datasets well

It is one of the most widely used ML algorithms.

🎯 Tasks for Day 41

🧠 Task 1: Theory

1. What is Random Forest?
2. What is an Ensemble Method?

3. What is Voting?
4. Advantages of Random Forest?
5. Difference between Decision Tree and Random Forest?

💻 Task 2: Dataset Creation

1. Student Result Dataset
2. Disease Dataset
3. Loan Approval Dataset

💻 Task 3: Model Building

1. Create Random Forest model
2. Train model
3. Make predictions

💻 Task 4: Prediction Practice

Predict:

1. Student Result
2. Disease Status
3. Loan Approval

💻 Task 5: Parameter Practice

Try:

n_estimators = 10
n_estimators = 50
n_estimators = 100

Compare outputs.

💻 Task 6: Compare Algorithms

Compare:

1. Decision Tree
2. Random Forest S
3. KNN

Write differences.

SS